→ Tutorial - 11

Dataset

sample	Feature1	Feature2	Feature3	Actual class	prediction
S_1	25	40	650	1	0.90
S_2	30	60	700	1	0.75
S_3	45	35	600	0	0.60
S_4	50	30	580	0	0.40
S_5	28	55	720	1	0.30

① Threshold value = 0.5 and compute the confusion matrix

② calculate the following matrics:

Accuracy

Precision

Recall (TPR)

F_1 - score.

⇒ Based on the threshold we will calculate the pred output.

Actual class	pred Prob	pred output
1	0.90	1 (TP)
1	0.75	1 (TP)
0	0.60	1 (FP)
0	0.40	0 (TN)
1	0.30	0 (FN)

class 1 = +ve
class 0 = -ve

Threshold value = 0.5

prob $\geq 0.5 \rightarrow$ class 1

prob $< 0.5 \rightarrow$ class 0

TP = True +ve

FP = False +ve

TN = True -ve

FN = False -ve

if donot get any one of the value then it is taken as '0'

confusion matrix.

prediction 1	TP 2	FP 1
prediction 0	FN 1	TN 1
	1	0

Actual

→ metrices

* precision = $\frac{TP}{TP+FP} = 0.66 \approx 0.67$

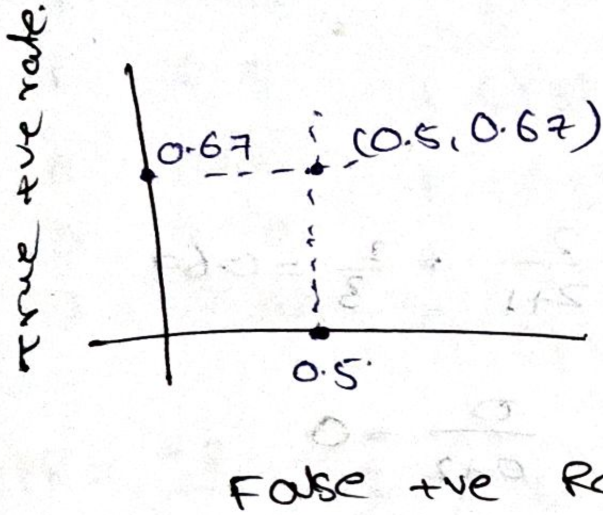
* Recall (TPR) = $\frac{TP}{TP+FN} = 0.67$

* F_1 score = $\frac{2 \times \text{precision}}{\text{precision} + \text{Recall}} = 0.67$

③ Calculate the FPR (False +ve Rate)

$$FPR = \frac{FP}{FP+TN} = 0.5$$

⑤ Plot the ROC



④ => For threshold => 0.7, 0.6, 0.5, 0.4, 0.3

=> Finding TPR and FPR

→ **For Threshold 0.7**

prob ≥ 0.7 → class ①
prob < 0.7 → class ⓪

Actual	pred	pred output
1	0.90	1 (TP)
1	0.75	1 (TP)
0	0.60	0 (TN)
0	0.40	0 (TN)
1	0.30	0 (FN)

confusion matrix.

Prediction \ Actual	1	0
1	TP 2	FP 0
0	FN 1	TN 2

→ $TPR = \frac{TP}{TP+FN} = \frac{2}{2+1} = \frac{2}{3} = 0.67$

→ $FPR = \frac{FP}{FP+TN} = \frac{0}{0+2} = 0$

⇒ **For Threshold 0.6**

prob ≥ 0.6 → class = 1

prob < 0.6 → class = 0

Actual	prob	prob output
1	0.90	1 (TP)
1	0.75	1 (TP)
0	0.60	1 (FP)
0	0.40	0 (TN)
1	0.30	0 (FN)

b) confusion matrix

Prediction \ Actual	1	0
1	TP 2	FP 1
0	FN 1	TN 1

→ $TPR = \frac{TP}{TP+FN} = \frac{2}{2+1} = \frac{2}{3} = 0.67$

→ $FPR = \frac{FP}{FP+TN} = \frac{1}{1+1} = \frac{1}{2} = 0.5$

→ **For threshold 0.4**

prob ≥ 0.4 → class 1

prob < 0.4 → class 0

Actual	prob	Prob Output
1	0.90	1 (TP)
1	0.75	1 (TP)
0	0.60	1 (FP)
0	0.40	1 (FP)
1	0.30	0 (FN)

Confusion matrix.

prediction	1	0
1	TP 2	FP 2
0	FN 1	TN 0

Actual

$$TPR = \frac{TP}{TP+FN} = \frac{2}{2+1} = \frac{2}{3} = 0.67$$

$$FPR = \frac{FP}{FP+TN} = \frac{2}{2+0} = \frac{2}{2} = 1$$

for threshold 0.3

prob ≥ 0.3 → class 1

prob < 0.3 → class 0

Actual	prob	prob output
1	0.90	1 (TP)
1	0.75	1 (TP)
0	0.60	1 (FP)
0	0.40	1 (FP)
1	0.30	1 (TP)

confusion matrix

prediction	1	0
1	TP 3	FP 2
0	FN. 0	TN 0

Actual

$$TPR = \frac{TP}{TP+FN} = \frac{3}{3+0} = \frac{3}{3} = 1$$

$\rightarrow FPR = \dfrac{FP}{FP+TN} = \dfrac{2}{2+0} = \dfrac{2}{2} = 1$

$\Rightarrow$ ROC curve for all thresholds

$\rightarrow$ 0.7, 0.5, 0.6, 0.4, 0.3